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COMPUTER ENGINEERING DEPARTMENT



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First Year Engineering Internship Projects Presentation

“ Restaurant Billing System “

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INTRODUCTION

Project Name: DineX

Tagline: Restaurant billing, simplified — bill orders in seconds, not spreadsheets.

Overview:

DineX is a lightweight, responsive web application designed specifically for small-to-medium restaurants, cafes, and eateries.

It bridges the gap between inefficient manual paper billing and overly complex, expensive POS (Point of Sale) enterprise software.

Provides seamless menu itemization, automated GST/tax computation, stock tracking, and instant bill/receipt generation.

OBJECTIVE

- Develop a secure Restaurant Billing Management System.
- Digitize the invoicing and receipt generation process.
- Reduce manual calculation errors and paperwork.
- Enable seamless administrator oversight of sales data.
- Provide secure staff authentication.
- Maintain centralized financial and sales records.

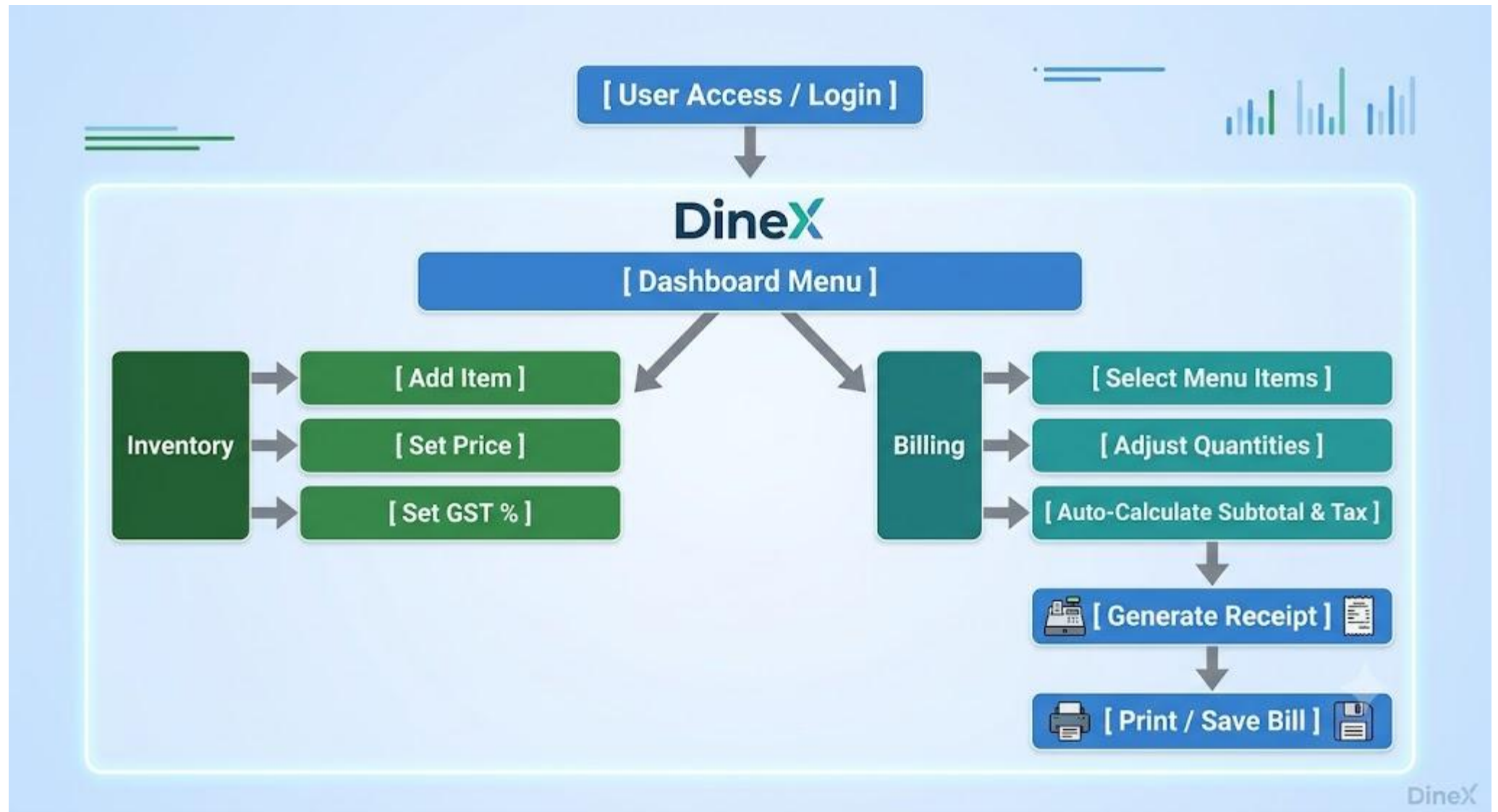
PROBLEM STATEMENT

Restaurants frequently face delays and errors due to manual billing processes. Calculating order totals, applying taxes, generating physical receipts, and tracking daily sales by hand is inefficient and can lead to financial discrepancies. Therefore, there is a need for a secure, centralized, and web-based billing platform that enables staff to generate quick, accurate invoices while allowing administrators to streamline financial records

MOTIVATION

- **Empowering Small Businesses:** Local diners and small restaurants need modern digital tools without the high subscription costs.
- **Speed & Efficiency:** Reducing customer wait times directly improves customer satisfaction and table turnover rates.
- **Digitalization:** Transitioning from spreadsheets to a dedicated Web POS improves business records and tax transparency.

FLOWCHART



SYSTEM DESIGN

- **Architecture:** Client-Side Web Architecture (Model-View Controller pattern / Component-based UI).
- **Modules:**
- **Authentication Module:** Manages user login, signup, and session access.
- **Inventory Management Module:** Serves as the menu database where items, prices, and GST percentages are defined.
- **Billing Engine:** Handles real-time shopping cart/order composition, subtotal calculation, and tax addition.
- **Print & Invoice Module:** Renders clean, print-ready digital receipts for physical printing or PDF export.

HARDWARE AND SOFTWARE REQUIREMENTS:

Hardware Requirements

- **Memory (RAM):** Minimum 2 GB (4 GB recommended).
- **Display:** 720p resolution or higher (Tablet, Laptop, or Desktop PC).
- **Peripheral:** Standard thermal receipt printer / desktop printer.
- **Internet connection.**

Software Requirements

- **Operating System:** Windows, macOS, Linux, Android, or iOS.
- **Web Browser:** Modern web browser (Google Chrome, Mozilla Firefox, Microsoft Edge, Safari).
- **Tech Stack:**
- **Frontend:** HTML5, CSS3, JavaScript (ES6+).
- **Deployment/Hosting:** GitHub Pages.

ALGORITHMS/METHODOLOGY:

1. Dynamic Tax Calculation Algorithm

$$\text{Item Total} = \text{Price} \times \text{Quantity}$$

$$\text{Tax Amount} = \text{Item Total} \times \left(\frac{\text{GST \%}}{100} \right)$$

$$\text{Grand Total} = \sum (\text{Item Total} + \text{Tax Amount})$$

2. Billing Methodology

- **Selection:** As items are added, an in-memory state (array/object) holds order items.
- **Real-time DOM Updates:** Event listeners trigger instant updates to the subtotal, GST sum, and final total whenever item quantity changes.
- **Receipt Formatting:** Applies CSS print media query rules (@media print) to format the output clean and ready for thermal or standard receipt printers.

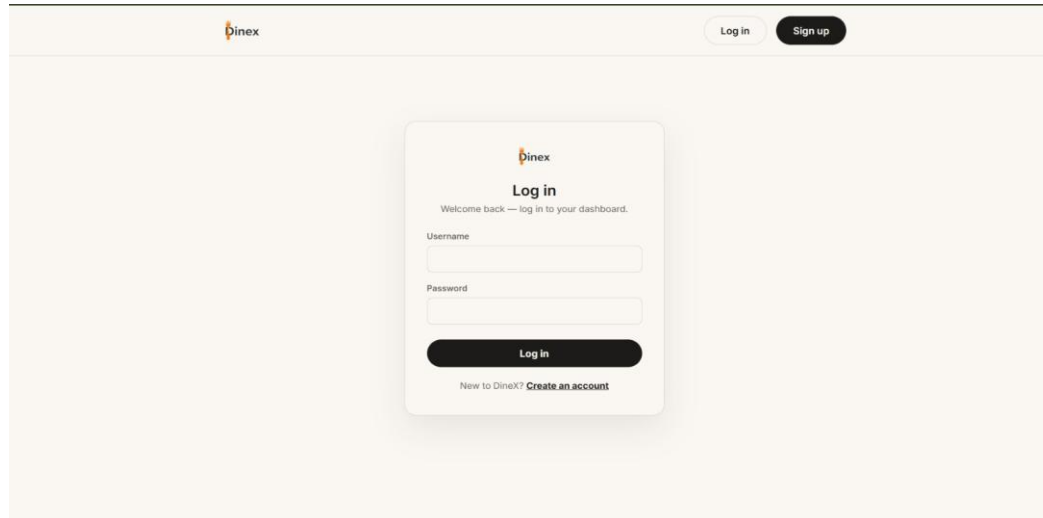
COMPARISON WITH EXISTING SYSTEMS :

Existing System	Dinex
Manual Billing	Automated Billing
Paper Records	Digital Records
Slow Process	Fast Process
More Errors	High Accuracy
Difficult Reports	Easy Report Generation
Manual Calculations	Automatic Calculations

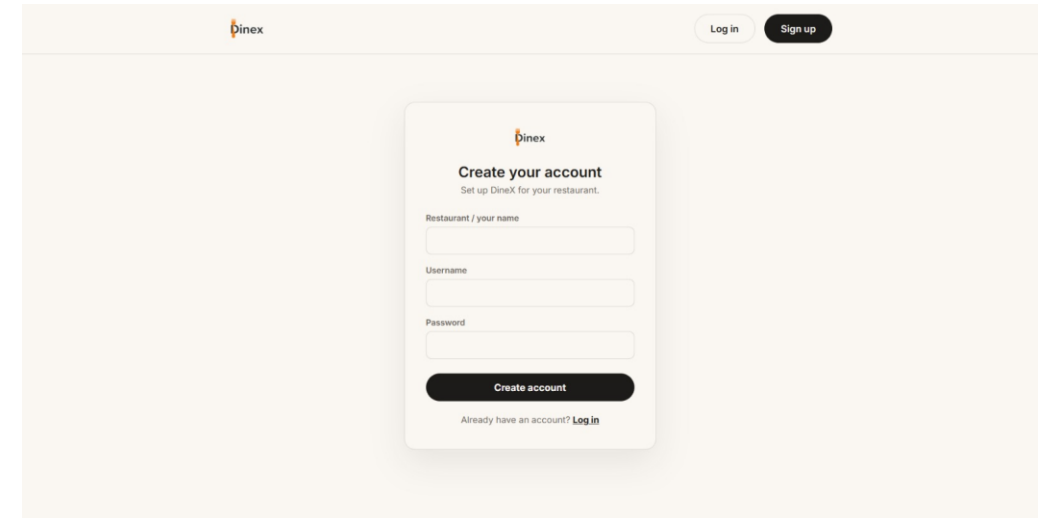
APPLICATIONS:

- **Restaurants** – Simplifies order management, billing, and customer service for dine-in and takeaway operations.
- **Cafes** – Enables quick order processing and efficient billing during peak hours.
- **Food Courts** – Manages multiple customer orders with fast and accurate billing.
- **Hotels** – Integrates restaurant billing with hotel dining services for seamless operations.
- **Fast Food Centers** – Speeds up order placement and billing to reduce customer waiting time.
- **Cloud Kitchens** – Organizes online food orders, billing, and delivery management efficiently.
- **College Canteens** – Handles high-volume food orders and cashless billing with ease.
- **Bakery Shops** – Manages product sales, inventory, and invoice generation for daily transactions.

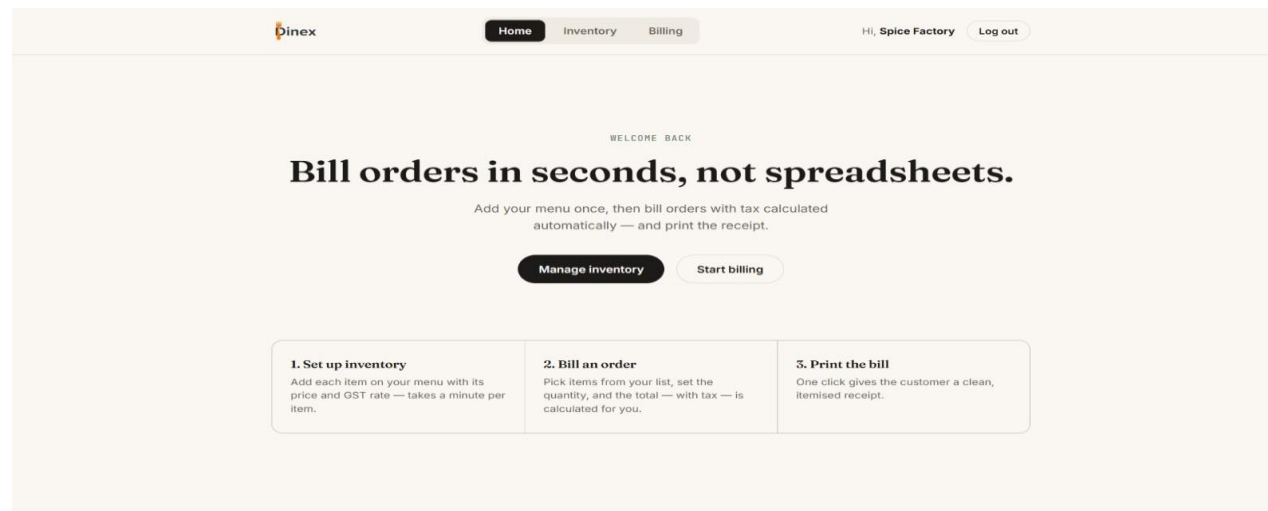
IMPLEMENTATION SNAPSHOTS:



The image shows the 'Log in' page of the Dinex application. At the top, there is a header with the Dinex logo on the left and 'Log in' and 'Sign up' buttons on the right. The main content area features a central white card with the Dinex logo, the title 'Log in', and a subtitle 'Welcome back — log in to your dashboard.' Below this, there are input fields for 'Username' and 'Password', followed by a 'Log in' button. At the bottom of the card, there is a link for 'New to DineX? Create an account'.



The image shows the 'Create your account' page of the Dinex application. At the top, there is a header with the Dinex logo on the left and 'Log in' and 'Sign up' buttons on the right. The main content area features a central white card with the Dinex logo, the title 'Create your account', and a subtitle 'Set up DineX for your restaurant.' Below this, there are input fields for 'Restaurant / your name', 'Username', and 'Password', followed by a 'Create account' button. At the bottom of the card, there is a link for 'Already have an account? Log in'.



The image shows the dashboard of the Dinex application. At the top, there is a header with the Dinex logo on the left, a navigation bar with 'Home', 'Inventory', and 'Billing' buttons, and a user profile section with 'Hi, Spice Factory' and a 'Log out' button. The main content area features a central white card with the title 'Bill orders in seconds, not spreadsheets.' and a subtitle 'Add your menu once, then bill orders with tax calculated automatically — and print the receipt.' Below this, there are two buttons: 'Manage inventory' and 'Start billing'. At the bottom, there is a table with three columns: '1. Set up inventory', '2. Bill an order', and '3. Print the bill'.

1. Set up inventory	2. Bill an order	3. Print the bill
Add each item on your menu with its price and GST rate — takes a minute per item.	Pick items from your list, set the quantity, and the total — with tax — is calculated for you.	One click gives the customer a clean, itemised receipt.

IMPLEMENTATION SNAPSHOTS:



Home

Inventory

Billing

Hi, Spice Factory

Log out

Inventory

Add the products your restaurant sells, with price and GST rate.

Product name

e.g. Margherita Pizza

Price (₹)

249

GST (%)

5

Add product

PRODUCT	PRICE	GST %	
Veg Platter	₹60.00	5%	Remove
Paneer Dum Tikka	₹20.00	5%	Remove
Paneer Cotton Kabab	₹20.00	5%	Remove
Cheese Cherry Pineapple	₹80.00	5%	Remove
Paneer Rajwadi	₹60.00	5%	Remove
Shahi Paneer	₹60.00	5%	Remove
Panner Mussallam	₹80.00	5%	Remove
Chicken Malwani Bone-less	₹99.00	5%	Remove
Mutton Nawabi Handi	₹1,200.00	5%	Remove

The screenshot displays the DineX mobile application's 'New Bill' interface. At the top, a navigation bar includes the DineX logo, a home icon, and links to 'Home', 'Inventory', 'Billing' (the active tab), 'Hi, Spice Factory', and 'Log out'. Below the navigation bar, a 'New Bill' section prompts the user to 'Tap a product to add it to the current bill.' The main content area is split into two panels. The left panel, titled 'New Bill', lists seven products with their prices and GST percentages, each accompanied by an 'Add' button: Butter Chicken (₹750.00 - GST 5%), Cheese Cherry Pineapple (₹380.00 - GST 5%), Chicken Angara (₹749.00 - GST 5%), Chicken Malwani Bone-less (₹999.00 - GST 5%), Mutton Biryani (₹849.00 - GST 5%), Mutton Nawabi Handi (₹1,280.00 - GST 5%), and Paneer Cotton Kabab (₹449.00 - GST 5%). The right panel, titled 'Table Bill', summarizes the current bill: Butter Chicken (₹750.00 × 1 = ₹750.50), Cheese Cherry Pineapple (₹380.00 × 1 = ₹399.80), a subtotal of ₹1,138.00, a GST of ₹56.50, and a final total of ₹1,186.50. At the bottom of the right panel are buttons for 'Print bill' and 'Clear bill'.

 dinex

Home

Inventory

Billing

Hi, Spice Factory

Log out

 dinex

Spice Factory

28 Jul 2026, 07:25 AM

Butter Chicken × 1	₹787.50
Cheese Cherry Pineapple × 1	₹399.00
Subtotal	₹1,130.00
GST	₹56.50
TOTAL	₹1,186.50

Use your browser's print option (**Ctrl** + **P**) or **Cmd** + **P**) to print this receipt.

[← Back to billing](#)

CONCLUSION:

- **Summary:** DineX successfully delivers a fast, error-free, and user-friendly billing experience for restaurant operators.
- **Impact:** Eliminates reliance on complex spreadsheets or costly POS hardware, making digital billing accessible to small businesses.
- **Future Enhancements:**
 - Integration with persistent databases (MongoDB/Firebase).
 - Mobile POS responsive app / PWA conversion.
 - Sales analytics dashboard and revenue reporting over time.
 - Integration with UPI and digital payment gateways.

REFERENCES:

- PHP Documentation — <https://www.php.net>
- MySQL Documentation — <https://dev.mysql>.
- Git Documentation — <https://git-scm.com>
- GitHub — <https://github.com>
- XAMPP — <https://www.apachefriends.org>
- Visual Studio Code — <https://code.visualstudio.com>



Thank You....!!!!